

# ADITYA THAKUR

AI Engineer | LLM Applications | RAG Pipelines | NLP

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## PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building and deploying LLM-based applications, Retrieval-Augmented Generation (RAG) pipelines, and NLP solutions. Proficient in LangChain, Hugging Face Transformers, Python, and vector databases (FAISS, Pinecone). Demonstrated ability to architect end-to-end AI systems from data ingestion through model deployment, achieving 85-90%+ answer relevancy benchmarks. Adept at prompt engineering, agentic workflows, and optimizing system accuracy and scalability using Docker and CI/CD practices.

## TECHNICAL SKILLS

**LLM & AI Frameworks:** LangChain, Hugging Face Transformers, OpenAI API, Prompt Engineering, Agentic Workflows

**ML / DL Frameworks:** TensorFlow, PyTorch, Scikit-Learn

**NLP & RAG:** Retrieval-Augmented Generation (RAG), Advanced RAG Techniques, Sentence Transformers, Text Chunking

**Vector Databases:** FAISS, Pinecone

**API&BACKEND:** FASTAPI

**Programming:** Python, SQL, HTML, CSS

**Deployment & DevOps:** Streamlit cloud, Docker, Git, GitHub, CI/CD (basic), Hugging Face Spaces

**Data Processing:** Pandas, NumPy, Exploratory Data Analysis (EDA)

## PROFESSIONAL EXPERIENCE

### Deep Learning & NLP Trainee | [Coder Roots](#)

June 2025 – July 2025

- Developed a production-ready PDF Q&A RAG application enabling users to upload documents and generate contextual question-answer pairs using LangChain, Streamlit, and Generative AI.
- Implemented complete RAG pipeline covering PDF text extraction, text chunking, embedding generation, and semantic retrieval with FAISS vector database.
- Utilized technologies: Python, LangChain, RAG, Generative AI, Streamlit, HTML, CSS.
- Deployed application publicly on Hugging Face Spaces, demonstrating end-to-end MLOps skills.

### Data Science & Machine Learning Trainee | [07 Services](#)

June 2024 – September 2024

- Built and deployed predictive CNN models and interactive data applications using Streamlit.
- Processed large datasets using Pandas and NumPy; conducted EDA and implemented regression, classification, and clustering models.
- Delivered an end-to-end ML cloth prediction application, demonstrating full project lifecycle execution.

## KEY PROJECTS

### PDF Q&A RAG Application | [LINK](#)

- Built an intelligent PDF Question-Answering system using LangChain, PyPDF, Hugging Face Transformers, and FAISS for document querying.
- Developed complete pipeline: PDF text extraction, chunking, embedding creation using sentence-transformers, and similarity-based semantic retrieval.
- Integrated open-source LLaMA models for accurate response generation; optimized retrieval via chunk-size tuning and metadata filtering.
- Achieved 85-90% relevance accuracy on user test cases; deployed with a clean Streamlit-based UI on Hugging Face Spaces.

YouTube Q&A RAG Application

- Developed a YouTube-based Retrieval-Augmented Generation (RAG) application for transcript-based question answering using LangChain, Streamlit, and Hugging Face Transformers.
- Implemented end-to-end pipeline: YouTube transcript extraction, text chunking, embedding generation via sentence-transformers, and semantic retrieval using FAISS vector database.
- Integrated open-source LLMs for contextual answer generation; achieved 90%+ answer relevancy on evaluation queries.
- Designed interactive Streamlit interface and deployed on Hugging Face Spaces for public access.

Indian Cybercrime Legal RAG Assistant |

- Built a Retrieval-Augmented Generation (RAG) application for Indian cyber law research, enabling users to describe situations in plain English and receive AI-generated legal analysis grounded in the IT Act 2000, IPC, and BNS 2023.
- Designed and curated a structured dataset of 153 real-world cybercrime scenarios across 12 categories (hacking, financial fraud, cyberstalking, data theft, etc.) with mapped legal sections, explanations, and punishments.
- Engineered a semantic search pipeline using FAISS vector indexing and Sentence-Transformers (all-MiniLM-L6-v2) embeddings to retrieve the most relevant legal provisions for any user query with sub-second latency.
- Implemented a LangChain-based RAG chain that retrieves context from the FAISS vector store, formats multi-field legal metadata, and generates grounded responses via Groq-hosted LLaMA-3.3-70B with temperature-controlled inference.
- Developed a conversational Streamlit interface with chat history, configurable retrieval depth (top-k), example query shortcuts, source transparency (showing retrieved entries with full metadata), and sidebar API key management.
- Ensured privacy-first architecture by running all embedding generation locally on-device, with no user data leaving the machine during indexing; only the final query is sent to the Groq LLM API for inferenc

EDUCATION

|                                                           |                     |
|-----------------------------------------------------------|---------------------|
| B.Tech in Computer Science & Engineering — DAV University | 2022 – 2026         |
| Senior Secondary (12th Grade) — S.D.S.V.M Talwara         | 74.6%   2021 – 2022 |
| Secondary (10th Grade) — S.D.S.V.M Talwara                | 79.8%   2018 – 2019 |